European Commission

ANNEX 1

Horizon Europe (HORIZON)

Description of the action (DoA)

Part A

Part B

DESCRIPTION OF THE ACTION (PART A)

COVER PAGE

Part A of the Description of the Action (DoA) must be completed directly on the Portal Grant Preparation screens.

PROJECT *Grant Preparation (General Information screen) — Enter the info.*	
Project number:	101168560
Project name:	A COMPREHENSIVE TRUSTWORTHY FRAMEWORK FOR CONNECTED MACHINE LEARNING AND SECURE INTERCONNECTED AI SOLUTIONS
Project acronym:	CoEvolution
Call:	HORIZON-CL3-2023-CS-01
Topic:	HORIZON-CL3-2023-CS-01-03
Type of action:	HORIZON-RIA
Service:	CNECT/H/01
Project starting date:	fixed date: 1 November 2024
Project duration:	36 months

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PROJECT SUMMARY

Project summary
Grant Preparation (General Information screen) — Provide an overall description of your project (including context and overall objectives, planned activities and main achievements, and expected results and impacts (on target groups, change procedures, capacities, innovation etc)). This summary should give readers a clear idea of what your project is about. *Use the project summary from your proposal.*
The contemporary AI landscape demands a holistic framework ensuring security across the supply chain and entire AI lifecycle. Despite existing adversarial attack techniques, a comprehensive end-to-end flow for identifying threats and vulnerabilities with associated risks is lacking. The EU, through initiatives like the AI Act, emphasizes safety and trustworthiness in AI applications but lacks a system managing weaknesses in a networked AI-supply chain. The CoEvolution project integrates its architecture components to create an end-to-end Security, Trust, and Robustness (STR) assessment solution, generating context-aware AI models characterized by their AI Model Bill of Materials (AIMBOM). The goal is a universal hub providing a coherent STR risk assessment and security assurance flow, aligning with MLDevOps and EU AI regulatory frameworks. The paradigm includes novel AI model descriptions, AIMBOM management, security monitoring, and context awareness. CoEvolution introduces a new STR paradigm based on Bills-of-Materials, offering a unified approach to describing AI models in supply chains, ensuring STR compliance with EU directives on trust, fairness, data governance, and GDPR guidelines. Open source trusted datasets and CoEvolution-developed AI models enhance the hub's capabilities, aiming for a robust, adaptable risk analysis and security assessment framework aligned with evolving AI cybersecurity threats.

LIST OF PARTICIPANTS

PARTICIPANTS

Grant Preparation (Beneficiaries screen) — Enter the info.

Number	Role	Short name	Legal name	Country	PIC
1	COO	ICCS	EREVNITIKO PANEPISTIMIAKO INSTITOUTO SYSTIMATON EPIKOINONION KAI YPOLOGISTON	EL	999654356
2	BEN	ISI	ATHINA-EREVNITIKO KENTRO KAINOTOMIAS STIS TECHNOLOGIES TIS PLIROFORIAS, TON EPIKOINONION KAI TIS GNOSIS	EL	999562788
3	BEN	UNICA	UNIVERSITA DEGLI STUDI DI CAGLIARI	IT	999841663
4	BEN	IMT	INSTITUT MINES-TELECOM	FR	999849326
5	BEN	SIE	SIEMENS AKTIENGESELLSCHAFT	DE	999987260
6	BEN	AEGIS	AEGIS IT RESEARCH GMBH	DE	911867804
7	BEN	TID	TELEFONICA INNOVACION DIGITAL SL	ES	917790915
8	BEN	UNIPI	UNIVERSITA DI PISA	IT	999862712
10	BEN	SLC	SECURITY LABS CONSULTING LIMITED	IE	895392645
11	BEN	AVL	AVL LIST GMBH	AT	999952243
12	BEN	AVI	AVISENSE.AI TECHNOVLASTOS IKE	EL	881486337
13	BEN	DIE	DIINEKES S.I. MONOPROSOPI IDIOTIKI KEFALAIOUCHIKI ETAIREIA	EL	884438338

PARTICIPANTS *Grant Preparation (Beneficiaries screen) — Enter the info.*					
Number	**Role**	**Short name**	**Legal name**	**Country**	**PIC**
14	BEN	CBRL	CYBERALYTICS LIMITED	CY	886488239
15	AP	STS	SPHYNX TECHNOLOGY SOLUTIONS AG	CH	881338703

LIST OF WORK PACKAGES

Work packages *Grant Preparation (Work Packages screen) — Enter the info.*						
Work Package No	**Work Package name**	**Lead Beneficiary**	**Effort (Person-Months)**	**Start Month**	**End Month**	**Deliverables**
WP1	Project Management	1 - ICCS	48.00	1	36	D1.1 – Project Manual D1.2 – Data Management Plan D1.3 – Innovation management and ethics oversight report D1.4 – Analysis of the legal and ethical issues
WP2	Dissemination, Exploitation, Sustainability	14 - CBRL	99.00	1	36	D2.1 – Plan for impact delivery D2.2 – First Impact Analysis D2.3 – Final Impact report
WP3	Specification/ Requirements and CoEvolution Architecture	2 - ISI	68.00	1	12	D3.1 – Project scope definition D3.2 – Technical specifications
WP4	AI model STR Assessment., Enhancement and Context Awareness technology enablers	7 - TID	210.00	8	33	D4.1 – Enablers development S&T report D4.2 – Enablers repository
WP5	CoEvolution AI model lifecycle STR Security Assurance and Monitoring	13 - DIE	161.00	8	33	D5.1 – Security assurance and monitoring S&T report D5.2 – Security assurance and monitoring repository
WP6	Integration, evaluation and project outcomes	12 - AVI	131.00	6	36	D6.1 – Simulator & datasets D6.2 – MVP D6.3 – Integrated framework D6.4 – Evaluation report

Work package WP1 – Project Management

Work Package Number	WP1	**Lead Beneficiary**	1 - ICCS
Work Package Name	Project Management		
Start Month	1	**End Month**	36

Objectives
The main objective of WP1 is to coordinate all management activities across the CoEvolution consortium. In particular, WP1 will address the following: • Set up the project coordination bodies; • Manage the project according to approved plans; • Monitor, track and control deviations due to progress, costs, financial and scheduling changes; • Implement procedures for quality management; • Implement an administration and communication infrastructure to establish a basis for efficient and easy communication within the project; • Perform a procedure for updating and revising the plans every 12 months due to changes and new knowledge; • Monitor and enact plans for gender equality; • European regulations compliance.

Description
Task 1.1 - Project Coordination and reporting; [Leader: ICCS; Participants: All; M01-M36]: This task is devoted to administrative and financial issues of the project and it comprises of i) project planning and scheduling, including tasks, responsibilities, and timescales; ii) consortium communication and project management such as adaptation and review of the work plan in line with monitoring feedback and organising regular teleconferences and face to face meetings; iii) reporting by collecting and consolidating partners' periodic reports, thus ensuring that cost claims for each partner and the consortium as a whole are duly justified and documented, and securing full compliance of EC Grant Agreement; iv) financial management, e.g., distribution of EU contribution among partners and financial project-wide records. Task 1.2 - Scientific and technical coordination, and monitoring of project's achievements; [Leader: ISI; Participants: -; M01-M36]: This task will produce a roadmap to be used as a guideline by the respective WPs to carry out their work. This task will be a) driving the technical activity; b) supporting the project coordinator in reporting all technical issues; c) arranging technical discussions and meetings with WP leaders; d) identifying technical risks and conflicts and taking corrective actions supporting the PC in risk management activities. Furthermore, a team will be formed to track and monitor the project's objectives and achievements and ensure that a methodology for measuring each KPI is in place. Task 1.3 - Quality assurance, gender equality assessment and risk management; [Leader: ICCS; Participants: TID; M01-M36]:This task will deal with all the activities necessary to promote gender equality within the frame of the project. Therefore, this task will actively encourage the employment of people from the spectrum of genders and groups in the project and will also engage the partners' gender equality committees at least twice in the project lifetime. Moreover, this task will ensure that all work conducted in the project meets the expected quality by defining an analytic Quality Management Plan, defining quality procedures and project standards. Finally, proactive strategies and contingency plans will be developed to mitigate and respond to identified risks (Table 5). Continuous monitoring, transparent communication with stakeholders, and regular reporting will ensure adaptability to changing project environments. Task 1.4 - Innovation tracking and data management; [Leader: AEGIS; Participants: -; M01-M36]:CoEvolution incorporates research, innovation strategy and monitoring as a core feature in its structure. In this context, this task aims at identifying and analysing the innovation potentials of CoEvolution and providing additional information and directions both to the project technical activities (with respect to current and emerging market needs) and to the project exploitation activities (with respect to exploitation plans and strategies). Innovation workshops may be organized as part of CoEvolution dissemination and exploitation strategy to invite potential adopters of CoEvolution results, where they will have a discussion playground. Moreover, it will establish open science and data management planning activities, which involve the identification of the relevant information that can be shared among the scientific communities for research purposes, and details the procedures for preserving and releasing project's open assets even beyond the end of the project. Task 1.5 - Ethical principles and European regulations, fundamental rights, and social sciences; [Leader: AEGIS; Participants: ISI, TID; M01-M36]:In this task, an Ethical Board (constituted by internal and external experts) will be established with the aim of assisting the consortium and the involved beneficiaries to quickly solve any uprising issues without affecting the project outcomes. Close monitoring and evaluation will be periodically performed in order to verify that each party is compliant with all European laws and regulations applicable to its activities, including the

acquisition of data, the processing of data through any tool used in connection with the project and the use of such data within the project framework. This task will also provide an analysis of the legal and ethical issues in the project (in particular focusing on privacy, data protection and AI regulation) to partners. More specifically, this task will monitor closely the EU approach on Artificial Intelligence, such as the proposed Artificial Intelligence Act and will keep the consortium apprised in order for CoEvolution to harmonize the results of the project with the EU guidelines. Finally, this task goes one step beyond by considering which legal issues may emerge for the CoEvolution outcomes in the post-project phase, and how the consortium can develop outcomes that are legally compliant and meet a high ethical standard.

Work package WP2 – Dissemination, Exploitation, Sustainability

Work Package Number	WP2	**Lead Beneficiary**	14 - CBRL
Work Package Name	Dissemination, Exploitation, Sustainability		
Start Month	1	**End Month**	36

Objectives
The main objective of this WP is to deliver the outcomes of CoEvolution to market/stakeholders and research community (Obj. 7). To this end, this work package will aim at: • Maximizing the impact of the project aligning business opportunities with the technical and research activity; • Assisting and complementing the technical development with the business perspective particularly, relating to future uptake and sustainability; • Studying the external context for CoEvolution results, providing input and requirements relating to market needs and trends and defining the market context for exploitation; • Ensuring proper communication of the project results and subsequently raising awareness to the scientific, industrial, and general public communities; • Following, contributing to, promoting and ensuring usage of the corresponding relevant standards, while also supporting the liaison and collaboration activities with other EC funded related projects and initiatives.

Description
Task 2.1 - Outreach planning, Communication and Dissemination; [Leader: AEGIS; Participants: All; M01- M36]:This task aims to establish a framework enabling partners to document their communication and dissemination efforts and assess their impact. These activities will be meticulously evaluated against the project's Key Performance Indicators (KPIs), and a flexible strategy will be formulated in real-time based on the extent to which these activities are fulfilled. This approach ensures the implementation of consistent communication initiatives across diverse channels, guaranteeing that pertinent stakeholders are well-informed about the project's progress. Special attention will be given to generating digital content and utilizing social media platforms such as Twitter and LinkedIn. Additionally, the project aims at enhancing its visibility and outreach with various tools and social media platorms, ensuring effective communication with its target audience. Task 2.2 - Standardisation activities; [Leader: CBRL; Participants: ISI, SIE, TID, AVL, STS; M01- M36]:The purpose of this task is to continuously investigate opportunities for CoEvolution to contribute to existing standards related to AI security and AI-enabled cybersecurity. After the careful examination of existing and relevant requirements from recognized organizations, this task will continuously investigate opportunities for CoEvolution to contribute to existing standards and provide relevant feedback to the standardization bodies and organizations. CoEvolution will devise a strategy and take necessary actions to try to influence the latter and the evolution of standards, by pushing the results of CoEvolution innovation activity. This task will also provide a set of best practices towards standardisation procedures. Task 2.3 - Liaison with other EU-funded projects and initiatives; [Leader: ISI; Participants: IMT, DIE, CBRL, STS; M01-M36]:This task involves actively engaging with other ongoing EU-funded projects and initiatives. As there are clearly links with privacy-preserving technologies and topics that are listed in "Increased Cybersecurity 2023", related projects are envisioned to be approached and identify opportunities for mutual learning and knowledge exchange, especially with proposals granted in HORIZON-CL3-2023-CS-01. Therefore, communication channels will be established with the aim of identifying potential areas for collaboration, and developing joint activities or outcomes. The liaison activities may include attending relevant meetings and events, conducting joint research, sharing updates on project developments, and establishing connections with project leaders and stakeholders. The aim is to ensure that the project is aligned with the broader goals of the EU's research and innovation agenda, and that opportunities for collaboration are maximised. Task 2.4 - Exploitation activities and long-term sustainability; [Leader: AVL; Participants: SIE, AEGIS, TID, SLC, AVL,

AVI, DIE, CBRL, STS; M08-M36]:Aiming at facilitating the sustainability and continuity of CoEvolution, this task will (i) investigate the market for the project on the short and long run (5 years after the project) by performing a rigorous, multi-stakeholder analysis on the total costs revealing the real costs of operating the CoEvolution framework on a day-to-day business; (ii) deliver a report on the exploitation activities carried out within the project's duration. Towards this direction, a suitable licensing strategy will be identified to ensure the commercial viability of the CoEvolution framework; (iii) Based on market potentials, a detailed business plan will be composed that includes SWOT analysis, market analysis, financial analysis, as well as marketing plans. This business plan will also elaborate on the value chains and business models that will allow the partners to commercially exploit the project's results. Task 2.5 - End user internal organization training activities; [Leader: SIE; Participants: AEGIS, AVL; M24-M36]: The objective of this Task is to ensure the uptake of project findings within the big organizations participating in the project. It will ensure dissemination of know-how across more departments than those involved in the project, seeking new applications in other departments and organisational processes. To implement the task, the project will develop training material and hold at least 1 workshop for each of the involved partners in which key-people from other departments (not involved in the project) will attend.

Work package WP3 – Specification/ Requirements and CoEvolution Architecture

Work Package Number	WP3	**Lead Beneficiary**	2 - ISI
Work Package Name	Specification/ Requirements and CoEvolution Architecture		
Start Month	1	**End Month**	12

Objectives
The objectives of this WP can be summarized in the following: • Conduct a detailed analysis of the state-of-the-art in AI, security, and cybersecurity; • Characterize potential security and privacy threats against networked AI systems; • Design, implement, and evaluate novel attacks and threat models collaboratively within the CoEvolution project; • Identify and continuously update the technical requirements; • Establish requirements and specifications for interoperability, security, and data integrity of networked AI systems within the CoEvolution project; • Define the architecture of the CoEvolution framework.

Description
Task 3.1 - State-of-the-Art, Threat model and Technical Specifications; [Leader: ICCS; Participants: ISI, UNICA, IMT, AEGIS, TID, UNIPI, SLC, DIE, CBRL; M01-M12]:This task will perform a detailed analysis of the state- of-the-art aimed at tracking the technological trends in the scope of the CoEvolution project, including research areas such as AI, security and cybersecurity. Relevant tools, processes, models and systems will be reviewed and presented in a living document, thus highlighting the way that the state-of-the-art meets the requirements defined in Task 3.2. Furthermore, a thorough characterization of the potential security and privacy threats against the networked AI systems will also be made available in the living document, existing threat models will be collected by studying the relevant literature and attack trends found in public threat reports, and novel attacks and threat models will be designed, implemented, and evaluated through the collaborative research efforts of the CoEvolution project. This will allow the CoEvolution framework to demonstrate its protective capabilities against the specific threat models. Finally, the technical requirements of the CoEvolution framework will be identified and will be continuously updated to be in line with the Use-Case specifications and requirements of Task 3.2. Task 3.2 - Use-Case Specifications and Requirements; [Leader: AVL; Participants: SIE, AEGIS, SLC, AVI; M01-M12]:This task is concerned with the collection of technological and industrial requirements and overall specifications for the proposed use cases. The requirements are to go beyond the specific sections of the AI supply chain and consider a holistic integration of the CoEvolution framework within the AI supply chain and automotive sector aimed at mitigating adversarial attacks, data poisoning and countering membership inference and evasion attacks to safeguard data privacy and model integrity, while also developing specific paradigms for context awareness. Special focus is dedicated to defining requirements and specifications for the interoperability, security, and data integrity of networked AI systems. Furthermore, AVL will introduce the process and tools for the continuous monitoring and updating of the use cases, maintaining the information in a "living" document until the final evaluation takes place. Task 3.3 - CoEvolution Reference Architecture; [Leader: ISI; Participants: ICCS, UNICA, IMT, AEGIS, UNIPI, CBRL;

M01-M12]:The activities carried out in this task include the definition of the CoEvolution framework's architecture. The architecture defined in this task serves as the main reference for further implementation, however, minor changes or adjustments may need to be performed. Within this task the different CoEvolution components, tools and engines will be analysed and be accommodated in the reference architecture, following a thorough understanding of the challenges, technologies, requirements, and state of the art introduced in the previous tasks of this WP. More specifically: (i) architectural components and functional blocks will be identified, such as the AI context awareness engine and STR Assurance engine; (ii) functional specifications stemming from T3.1, T3.2 and T3.4 will be interpreted; (iii) integration interfaces between the various components and engines will be defined and trustworthy communications will be ensured; (iv) in case there are variations in the architecture for the different pilot scenarios, they will be noted; (v) information flows and control flows between the different layers; (vi) an initial roadmap on the integration and deployment process that will be the basis for T6.1. The developed architecture and functional specification will drive the work of WP4 and WP5. The architecture will be continuously updated following the technical and business achievements of the project.
Task 3.4 - Data requirements and collection; [Leader: DIE; Participants: ISI, AEGIS, SLC, DIE; M01-M12]:The entire lifecycle of AI models from design and training to deployment is highly dependent on data and that is the case for the research carried out in the CoEvolution project. The tasks within WPs 3-5 are going to develop mainly data-driven models and monitoring engines. As such, it is imperative to understand the data requirements and strive early in the project to collect all the necessary data sources. At the same time, the exploitation of the data that will be extracted through the project processes needs to be carefully planned and monitored. This task will investigate open source secure datasets, such as the Comprehensive, Multi-Source Cyber-Security Events, the DARPA Intrusion Detection Data Sets, while also provide requirement for the kind of data the use-case specific Data Generation Mechanism, the CoEvolution simulator, should be required to generate, which will be developed in T6.2 in order to generate useful data which will be considered trusted.

Work package WP4 – AI model STR Assessment., Enhancement and Context Awareness technology enablers

Work Package Number	WP4	**Lead Beneficiary**	7 - TID
Work Package Name	AI model STR Assessment., Enhancement and Context Awareness technology enablers		
Start Month	8	**End Month**	33

Objectives
The work carried out in this WP focuses on creating a set of tools, services, and methodologies to enhance the security and resilience of AI models during the entire lifecycle of AI models. The result will be a secure design/training/deployment pipeline with protection against adversarial attacks. The objectives of this WP can be summarized as follows: • Provide a toolbox for secure and resilient AI model design, training and deployment; • Design a framework through which AI models can safeguard against adversarial attacks based on the context; • Incrementally learn novel vulnerability attacks and enhance the robustness and security of deployed AI models. • Implement add-on gadgets to accompany AI models that perform a number of functions, from telemetry and attack mitigation, to update mechanisms

Description
Task 4.1 - Adversarial Attack Security Testing/Assessment techniques; [Leader: AVI; Participants: ICCS, ISI, UNICA, TID, UNIPI, IMT, SIE, AVL; M08-M33]: The goal of this task is to develop a series of adversarial attack techniques that will indicate the exposure of AI models to various adversarial attack categories, such as black/gray/white box attacks and methods such as model replacement, data poisoning and membership inference among others. The adversarial attack countermeasures at this phase aim to prepare the ground for the research work to be done in the design/training/deployment phases of the AI model lifecycle and introduce the building block for adversarial attack resistance at the training and deployment phases. The solutions provided at this stage since we are also considering collaborative learning and AI interconnected, full AI supply chain agents, are also focused on structuring the proper interconnections between AI models that can guarantee security and privacy. This task will also develop methodologies and metrics in order to assess the "safety" of datasets, i.e. whether they have been poisoned or not or they have bias or not. More specifically, this task will also assess dataset issues such as data bias, anonymity/privacy of training data as well as possible adversarial poisoning on the training/testing/verification datasets. This task will produce a series of security assessment tools based on adversarial attack vectors acting as data

evaluation mechanism and an AI model fuzzer as well as a data privacy assessment tool. Finally, this task will deal with information leakage for AI models deployed in computation infrastructure by developing side channel leakage assessment tools to identify the operations that are performed and infer the model structure. The assessment tools and techniques of this task will be used for T5.3 to structure the CoEvolution STR security Testing/Assessment Engine.

Task 4.2 - Design and Training Phase Defences Technologies; [Leader: UNICA; Participants: ICCS, ISI, TID, UNIPI, IMT, SIE, AVL, AVI; M08-M33]:

This task will develop a set of tools and services with the aim of providing a training pipeline with improved security, including protection against poisoning and backdoor attacks that tamper with the training process by manipulating either a fraction of the training data, or the learning algorithm and the model architecture (e.g., supply-chain attacks staged when model training is outsourced to an untrusted third party). In particular, we will develop protection mechanisms against data and model poisoning attacks (including backdoors), while keeping into account that models are continuously updated and deployed. To tackle this issue, we will characterise a tradeoff between model adaptability and vulnerability to poisoning in continual learning settings. We will also evaluate eXplainable AI (XAI) methods to design a debugging tool that can be used during model training to foster interpretability and transparency, while also enabling prompt detection of poisoning and backdoor attacks. Furthermore, collaborative AI methodologies will be integrated into the process to ensure fairness, with special attention to biases introduced by personal data. Finally, this task will work in close collaboration with the CoEvolution simulator of T6.2 to make sure that trusted data are used during the training phase.

Task 4.3 - Deployment Phase Defences Technologies; [Leader: TID; Participants: ICCS, ISI, UNICA, UNIPI, IMT, SIE, AVL, AVI; M08-M33]:

This task will focus on developing a set of gadgets, that will accompany an AI model during the deployment phase, ensuring that the deployed model or its results remain untampered, resist inference adversarial attacks, and avoid becoming unresponsive. This task will work in collaboration with T4.4 and T4.5 and the deployed models will be infused with gadgets that provide security and context self-awareness. More specifically, these AI gadgets are designed with three possible capabilities: a) to detect an inference-based adversarial attack on a model and report it as an incident (security self-awareness), b) to update their defenses when new attacks are discovered by the AI security research community, and c) to run in an execution environment that can be considered secure and efficient, thus minimizing risks associated with tainted open-source models and datasets. Furthermore, monitoring mechanisms will be developed to provide telemetry track and report possible malicious attacks to the deployed models such as the last layer attack, GPU overflow, or tainted open-source model deployments and provide mitigating actions. The result of this task is a resilient AI model, equipped with tools, capable of preventing, detecting, and mitigating adversarial attacks in real-world scenarios. This includes considering the specifications and constraints of the deployment system, as well as potential exploits within the execution environment that could be leveraged to mount an adversarial attack.

Task 4.4 - AI Model Context-Awareness Enablers for Security, Robustness and Performance; [Leader: TID; Participants: IMT; M08-M33]:

This task is dedicated to researching, designing, and developing a technological framework for context-aware AI resilience by means of data sensing, storing, and dynamically updating and evaluating context in distributed FL and decentralised P2PL architectures. More specifically, T4.4 will evaluate the utility of context in a decision support framework within FL and P2P learning scenarios. Context vectors, enriched with real-time data acquired from the operating environment will serve as the observational input for the context decision support agent (CDS), which will be trained to maximise objectives related to privacy and security. The decision support agent will use the context vectors to learn the best policy about protocol adaptations and other operational parameters. The agent's objective will be two-fold: (1) to maximise the privacy of data exchanges, and (2) to fortify the security protocols in place. The decision support agent will dynamically adjust its policy based on context, implementing optimal actions such as (but not limited to) adapting communication protocols to mitigate the risk of data breaches or eavesdropping, strategically connecting with trustworthy users, thereby reinforcing the reliability of the learning network, and adapting the frequency and granularity of model updates shared across the network to balance privacy with collaborative learning efficacy.

Task 4.5 - Continuous Post-Deployment Defence through Incremental training technologies [Leader: UNIPI; Participants: ISI, UNICA, SIE, AVI, DIE; M08-M33]

This task aims to develop novel training techniques that enhance the security and robustness of AI models in an incremental manner, as and when new vulnerabilities are discovered. We propose novel methods to convert newly discovered vulnerabilities into security patches that can be applied through an additional training step. These patches can take various forms, such as a set of samples to forget, a synthetic dataset designed for privacy and robustness, or model parameters that must be combined with the original model. We will apply these patches using custom training algorithms that combine recent research in continual learning with security enhancement. The patch and training methods will be evaluated to ensure a good trade-off between the security improvements and the gap in accuracy compared to the unsafe models. The methods will improve the security of AI models by allowing them to patch them efficiently against novel vulnerabilities, or after changes in the threat model. In essence, this task will develop the format, the methods and also develop and implement the mechanism, a gadget, to support the actual enforcement of the patches as an update mechanism on the deployed AI model.

Work package WP5 – CoEvolution AI model lifecycle STR Security Assurance and Monitoring

Work Package Number	WP5	**Lead Beneficiary**	13 - DIE
Work Package Name	CoEvolution AI model lifecycle STR Security Assurance and Monitoring		
Start Month	8	**End Month**	33

Objectives
The objectives of this WP can be summarized in the following: • Develop the novel AI MBOM structure that is crucial to CoEvolution's operations; • Develop the Security Runtime Monitoring engine; • Develop the security and risk assessment engine; • Develop the CoEvolution Security Assurance Engine; • Develop context-awareness adaptation techniques.

Description
Task 5.1 - STR AI Model Bills-of-Material Definition, Design and Management; [Leader: ISI; Participants: AEGIS, UNIPI, STS; M08-M33]: The goal of this task is adopt the successful principles of software design and DevOps - the Bill-of-Materials - to provide detailed information about the model's training data, the owner and users, dependencies to external libraries, hardware used and the model's specific defence flow that CoEvolution will offer, including security and risk assessment scores, gadgets and enhancements. This comprehensive record-keeping ensures that AI models are thoroughly vetted, and their development process is transparent and accountable. The result of this task will be a novel comprehensive structure, called the AI MBOM, that encapsulates the core aspects of AI model design and development. This structure goes one step further by incorporating essential elements of security, trust, privacy, robustness, and explainability. AI MBOM's ability to update records with new data enables AI systems to remain up-to-date and adaptable to evolving environments. Its purpose is to instil confidence in users of AI models, whether in collaborative learning environments or as integral components of AI model supply chains as well as manage user's access on the AI MBOM, via authentication and authorization and integrity mechanisms. Task 5.2 - Security Runtime Monitoring; [Leader: DIE; Participants: ICCS, ISI, AEGIS, AVI, DIE; M08-M33]: The goal of this task is to develop the Security Runtime Monitoring engine that will offer adversarial cyberattack surveillance of AI operations. The engine will focus on the run-time operation of the AI models, more specifically monitoring the AI model collecting events (e.g adversarial attack incidents) from the monitoring and telemetry gadgets to identify inference adversarial attacks, as well as, validate, authenticate and continuously update the AI MBOM with information and events from the runtime operation of the model. In practice, the SRME is acting as an AI MBOM management trusted authority that is entitled to evaluate the validity of an AI MBOM (i.e check that all AI MBOM fields are valid) and to assess the AI MBOM integrity and authenticity. Therefore this engine will work closely with the tools and services developed in WP4 to increase its reliability and robustness. Task 5.3 - Risk Assessment Analysis and Security Assessment Engines; [Leader: SLC; Participants: ICCS, ISI, SLC, DIE; M08-M33]: This task will develop a security and risk assessment and guidelines/recommendation engine with the aim of securing the overall AI lifecycle from model design to model deployment. The engine will use tools and services from T4.1 and employ a holistic approach to risk management, considering potential cybersecurity threats and AI-related attacks while at the same time be compliant with legislations like the GDPR and the AI Act, with input from T1.5. The risk assessment and guidelines/recommendation engine will draw data from AI supply chain databases such as AI Risk Database and MITRE ATLAS and perform risk analysis on a given AI model to generate risk assessment based on the deployment scenario, as well as guidelines for detected vulnerabilities. The Security assessment engine will offer techniques and tools for mitigating detected vulnerabilities as well as generate the Adversarial Attack metrics that comprise the security assessment score, based on detected and mitigated vulnerabilities. Task 5.4 - Continuous Security Assurance and Trusted AI MBOM Generation Mechanism; [Leader: CBRL; Participants: ISI, UNIPI, DIE, STS; M08-M33]: This task will be responsible for the development of the CoEvolution Security Assurance Engine that serves as the orchestrating element of the CoEvolution Hub with respect to Risk Analysis, Security Assurance, Robustness and context awareness. The purpose of this engine is to ensure the alignment with AI trustworthiness and fairness guidelines and standards reported from T1.5, gather reports from the rest of the CoEvolution components and coordinate them.

Task 5.5 - Evolutionary strategies for context-awareness adaptation; [Leader: TID; Participants: UNICA, AEGIS; M08-M33]:
This task will focus on researching novel evolutionary strategies for facilitating a dynamic, context-aware AI adaptation for improved resilience against adversarial threats. To this end, T5.5 will develop genomic operations that involve generating a diverse population of context vectors, each undergoing operations such as crossover, mutation, and the creation of offspring. The resulting offspring from these genetic operations will be evaluated using relevant fitness functions that are closely tied to resilience and performance metrics. This strategy will ensure that only the most robust and efficient context vectors are selected for propagation to the subsequent generations, fostering a continuous cycle of improvement and adaptation. This guarantees that only the most robust and efficient context vectors are selected for propagation to the subsequent generations, fostering a continuous cycle of improvement and adaptation, to better suit the dynamic nature of the envisioned operating environments.

Work package WP6 – Integration, evaluation and project outcomes

Work Package Number	WP6	Lead Beneficiary	12 - AVI
Work Package Name	Integration, evaluation and project outcomes		
Start Month	6	End Month	36

Objectives

The objectives of this task can be summarized in the following:
- Integration and implementation of the CoEvolution framework;
- Develop a simulator for synthetic data creation and evaluation;
- Pilot deployment, testing and demonstration;
- Validation and evaluation of the CoEvolution pilots;
- Report project outcomes.

Description

Task 6.1 - CoEvolution hub Integration and Implementation; [Leader: AEGIS; Participants: All; M19-M24]:
In this task, the envisioned CoEvolution hub will be integrated and implemented, continuing the work of WP4 and WP5. Activities in this task will involve the following: i) explain which underlying domain data models will be used and why; ii) prescribe the necessary infrastructure; iii) assign responsibilities regarding the provision and installation of the necessary infrastructure for the effective system operation; iv) coordinate and integrate the tools and services developed in WPs 4 and 5; v) implement the CoEvolution hub.
In addition to the above, AVL and AVI will perform adaptations to AVL's testbed (vehicle and equipment) to ensure its compatibility with the prescribed infrastructure, tools and services leading to effective integration and implementation of the CoEvolution hub and enhanced availability of data sharing. This effort will ensure the seamless integration of AVL's vehicle and equipment into the CoEvolution hub.
Task 6.2 - Simulation for evaluation and dataset creation; [Leader: AVI; Participants: ICCS, IMT, SIE; M06-M24]:
The goal of this task is to develop an all-inclusive adversarial data generation mechanism, the Data Generation, which will be a simulator that will be exploited in the training phase through the use of synthetic data and in the evaluation phase. For the former, raw sensory data that are considered trusted can be generated and fed to the AI models. When malicious data are detected they will be removed and the simulator will create synthetic trusted data in order to fill out the data gap for the training phase. For the evaluation phase, virtual environments will be created and deployed for evaluating the generated AI models. To develop the Data Generator we will utilize the latest and novel techniques for creating perturbated data from existing datasets, such as using Generative Learning techniques, while focusing on attacks like Adversarial sample Generation and Model Inversion attacks.
Task 6.3 - Pilot Deployment, Testing and Demonstration; [Leader: SIE; Participants: All; M24-M30]:
Using an instance of the implemented CoEvolution framework from T6.1, this Task will deal with the deployment of the pilot prototypes for each use case that will serve as proofs of concept, by mapping the use case requirements against the developed components and engines of WP4 and WP5. Based on the definitions of the initial use cases from T3.2 the pilot prototypes will be tested and demonstration workshops with all partners of the consortium will take place to make sure that the needs of the end users and the respective KPIs have been met. All the use case and evaluation activities will be supervised by the WPL and the TC of the project, guiding, assisting, and resolving any issues related to the technical dimensions of WP6. Finally, this task will receive the attack scenarios of T3.1 and will define the evaluation

methodology, trials timing and planning, the specific KPIs and the testing protocol to be followed. This task will provide input to T6.4.
Some additional effort will be introduced by AVL and AVI in order to facilitate the transition of AVL's equipment maturity level, by introducing more vision sensors and the underlying systems to support data aggregation and processing.
Task 6.4 - Validation, Evaluation and project outcomes; [Leader: SLC; Participants: AVL, AVI, DIE, STS; M30-M36]:
The first part of this task relates to the pilot validation of the project. The CoEvolution framework will be validated under realistic conditions to reach the overall CoEvolution goals. End-users committed to supporting the project during pilot planning & preparation (T6.3) will execute pilot studies under the guidance of the task leader based on the pilot scenarios specified in Task 3.2. The task will involve the organization of events, workshops and other activities that will provide opportunities for stakeholders to use the CoEvolution set of technologies and provide feedback. Once the execution phase is over, the task will evaluate the pilot, summarize and report the outcomes and possible impacts of the CoEvolution framework for project partners, stakeholders, and beneficiaries in short term as well as in the long run. The task will further: i) summarize all intended and unintended outcomes of the pilot operations; ii) produce repositories for pilot scenario data and reports; iii) evaluate the CoEvolution framework from all stakeholders perspectives; iv) produce a range of best practices for the use of CoEvolution; v) and report future possibilities and solutions for the use of the CoEvolution framework.
Additional effort will be placed by AVL for providing a realistic simulation environment for the validation and demonstration of some use cases.

STAFF EFFORT

Staff effort per participant *Grant Preparation (Work packages - Effort screen) — Enter the info.*							
Participant	**WP1**	**WP2**	**WP3**	**WP4**	**WP5**	**WP6**	**Total Person-Months**
1 - ICCS	20.00	2.00	8.00	24.00	16.00	12.00	82.00
2 - ISI	8.00	8.00	10.00	32.00	21.00	5.00	84.00
3 - UNICA	1.00	2.00	4.00	33.00	8.00	4.00	52.00
4 - IMT	1.00	4.00	3.00	16.00		10.00	34.00
5 - SIE	1.00	9.00	3.00	10.00		15.00	38.00
6 - AEGIS	7.00	21.00	10.00		12.00	18.00	68.00
7 - TID	3.00	5.00	2.00	33.00	20.00	4.00	67.00
8 - UNIPI	1.00	2.00	3.00	28.00	18.00	4.00	56.00
10 - SLC	1.00	4.00	4.00		16.00	9.00	34.00
11 - AVL	1.00	12.00	6.00	6.00		14.00	39.00
12 - AVI	1.00	4.00	3.00	24.00	6.00	20.00	58.00
13 - DIE	1.00	6.00	6.00	4.00	20.00	7.00	44.00
14 - CBRL	1.00	12.00	4.00		14.00	4.00	35.00
15 - STS	1.00	8.00	2.00		10.00	5.00	26.00
Total Person-Months	48.00	99.00	68.00	210.00	161.00	131.00	717.00

LIST OF DELIVERABLES

Deliverables

Grant Preparation (Deliverables screen) — Enter the info.

The labels used mean:

Public — fully open (⚠ automatically posted online)
Sensitive — limited under the conditions of the Grant Agreement
EU classified —RESTREINT-UE/EU-RESTRICTED, CONFIDENTIEL-UE/EU-CONFIDENTIAL, SECRET-UE/EU-SECRET under Decision 2015/444

Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D1.1	Project Manual	WP1	1 - ICCS	R — Document, report	PU - Public	3
D1.2	Data Management Plan	WP1	6 - AEGIS	DMP — Data Management Plan	SEN - Sensitive	6
D1.3	Innovation management and ethics oversight report	WP1	6 - AEGIS	R — Document, report	SEN - Sensitive	36
D1.4	Analysis of the legal and ethical issues	WP1	6 - AEGIS	R — Document, report	PU - Public	12
D2.1	Plan for impact delivery	WP2	6 - AEGIS	R — Document, report	SEN - Sensitive	6
D2.2	First Impact Analysis	WP2	14 - CBRL	R — Document, report	SEN - Sensitive	18
D2.3	Final Impact report	WP2	14 - CBRL	R — Document, report	SEN - Sensitive	36
D3.1	Project scope definition	WP3	1 - ICCS	R — Document, report	PU - Public	12
D3.2	Technical specifications	WP3	2 - ISI	OTHER	PU - Public	12
D4.1	Enablers development S&T report	WP4	3 - UNICA	R — Document, report	PU - Public	18
D4.2	Enablers repository	WP4	7 - TID	OTHER	PU - Public	33
D5.1	Security assurance and monitoring S&T report	WP5	2 - ISI	R — Document, report	PU - Public	18

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D5.2	Security assurance and monitoring repository	WP5	13 - DIE	OTHER	PU - Public	33
D6.1	Simulator & datasets	WP6	12 - AVI	OTHER	PU - Public	24
D6.2	MVP	WP6	4 - IMT	OTHER	PU - Public	24
D6.3	Integrated framework	WP6	5 - SIE	DEM — Demonstrator, pilot, prototype	PU - Public	36
D6.4	Evaluation report	WP6	10 - SLC	R — Document, report	PU - Public	36

Deliverable D1.1 – Project Manual

Deliverable Number	D1.1	**Lead Beneficiary**	1 - ICCS
Deliverable Name	Project Manual		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	3	**Work Package No**	WP1

Description
Project processes for day-to-day operations and quality management

Deliverable D1.2 – Data Management Plan

Deliverable Number	D1.2	**Lead Beneficiary**	6 - AEGIS
Deliverable Name	Data Management Plan		
Type	DMP — Data Management Plan	**Dissemination Level**	SEN - Sensitive
Due Date (month)	6	**Work Package No**	WP1

Description
Data management plan and a plan for legal and ethical requirements analysis (live document)

Deliverable D1.3 – Innovation management and ethics oversight report

Deliverable Number	D1.3	**Lead Beneficiary**	6 - AEGIS
Deliverable Name	Innovation management and ethics oversight report		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP1

Description
Innovation management and ethics oversight report

Deliverable D1.4 – Analysis of the legal and ethical issues

Deliverable Number	D1.4	**Lead Beneficiary**	6 - AEGIS
Deliverable Name	Analysis of the legal and ethical issues		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	12	**Work Package No**	WP1

Description
Artificial Intelligence Act analysis and Ethical Board activities report (living document)

Deliverable D2.1 – Plan for impact delivery

Deliverable Number	D2.1	**Lead Beneficiary**	6 - AEGIS
Deliverable Name	Plan for impact delivery		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	6	**Work Package No**	WP2

Description
Communication, Dissemination and Exploitation Plan

Deliverable D2.2 – First Impact Analysis

Deliverable Number	D2.2	**Lead Beneficiary**	14 - CBRL
Deliverable Name	First Impact Analysis		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	18	**Work Package No**	WP2

Description
First report on the activities and indications on their impact

Deliverable D2.3 – Final Impact report

Deliverable Number	D2.3	**Lead Beneficiary**	14 - CBRL
Deliverable Name	Final Impact report		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP2

Description
Second report on the activities and indications on their impact

Deliverable D3.1 – Project scope definition

Deliverable Number	D3.1	**Lead Beneficiary**	1 - ICCS
Deliverable Name	Project scope definition		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	12	**Work Package No**	WP3

Description
State of the art report, threat models, codified requirements and use cases

Deliverable D3.2 – Technical specifications

Deliverable Number	D3.2	**Lead Beneficiary**	2 - ISI
Deliverable Name	Technical specifications		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	12	**Work Package No**	WP3

Description
Architecture and technical specification of the framework (live document)

Deliverable D4.1 – Enablers development S&T report

Deliverable Number	D4.1	**Lead Beneficiary**	3 - UNICA
Deliverable Name	Enablers development S&T report		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP4

Description
Scientific and technical report of the developed tools and techniques

Deliverable D4.2 – Enablers repository

Deliverable Number	D4.2	**Lead Beneficiary**	7 - TID
Deliverable Name	Enablers repository		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	33	**Work Package No**	WP4

Description
SW repository with developed tools with proper documentation for prospective uptakers

Deliverable D5.1 – Security assurance and monitoring S&T report

Deliverable Number	D5.1	**Lead Beneficiary**	2 - ISI
Deliverable Name	Security assurance and monitoring S&T report		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP5

Description
Scientific and technical report of the developed tools and techniques

Deliverable D5.2 – Security assurance and monitoring repository

Deliverable Number	D5.2	**Lead Beneficiary**	13 - DIE
Deliverable Name	Security assurance and monitoring repository		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	33	**Work Package No**	WP5

Description
SW repository with developed tools with proper documentation for prospective uptakers

Deliverable D6.1 – Simulator & datasets

Deliverable Number	D6.1	**Lead Beneficiary**	12 - AVI
Deliverable Name	Simulator & datasets		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	24	**Work Package No**	WP6

Description
Simulator and datasets to support R&D

Deliverable D6.2 – MVP

Deliverable Number	D6.2	**Lead Beneficiary**	4 - IMT
Deliverable Name	MVP		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	24	**Work Package No**	WP6

Description
A minimum viable product of the integrated framework to be validated and demonstrated

Deliverable D6.3 – Integrated framework

Deliverable Number	D6.3	**Lead Beneficiary**	5 - SIE
Deliverable Name	Integrated framework		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP6

Description
Final integrated framework and prototypes

Deliverable D6.4 – Evaluation report

Deliverable Number	D6.4	**Lead Beneficiary**	10 - SLC
Deliverable Name	Evaluation report		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP6

Description
Evalution process summary results for uptake and post-implementation support & development

LIST OF MILESTONES

Milestones *Grant Preparation (Milestones screen) — Enter the info.*					
Milestone No	**Milestone Name**	**Work Package No**	**Lead Beneficiary**	**Means of Verification**	**Due Date (month)**
1	Project bootstrapping	WP2, WP1	6 - AEGIS	D1.1,D1.2&D2.1	6
2	Project scope codification	WP3, WP1	1 - ICCS	D3.1,D3.2&D1.4	12
3	R&D and impact maximization plans and first results	WP2, WP4, WP5	2 - ISI	D2.2,D4.1&D5.1	18
4	MVP release	WP6	12 - AVI	D6.1&D6.2	24
5	S&T outputs for uptake	WP4, WP5	7 - TID	D4.2&D5.2	33
6	Project outputs, evaluation and sustainability	WP6, WP2, WP1	14 - CBRL	D1.3,D2.3,D6.3&D6.4	36

LIST OF CRITICAL RISKS

Critical risks & risk management strategy *Grant Preparation (Critical Risks screen) — Enter the info.*			
Risk number	**Description**	**Work Package No(s)**	**Proposed Mitigation Measures**
1	Partner leaving the project. Allocated and possibly key-part of the work cannot be performed, delaying the project schedule. (low/medium)	WP6, WP3, WP2, WP1, WP4, WP5	If possible, the affected tasks can be allocated to another partner(s). This is permitted by the consortium complementarity in terms of expertise. Alternatively, the management will perform an amendment to include a replacement partner. Especially regarding the UK, under the Withdrawal Agreement, the UK will continue to participate in programmes funded under H2020 until their closure.

Critical risks & risk management strategy *Grant Preparation (Critical Risks screen) — Enter the info.*			
Risk number	**Description**	**Work Package No(s)**	**Proposed Mitigation Measures**
2	Key-person left or is temporarily not available or part of work allocated to one partner cannot be performed on time and with expected quality. (low/ medium)	WP6, WP3, WP2, WP1, WP4, WP5	Consortium partners are involved in the related areas with more than one staff member, ensuring an immediate substitution. Furthermore, the project as a whole has technical excellence in related disciplines spread across the partners, providing additional substitution possibilities or an appropriate shift of resources and responsibilities.
3	Resources are underestimated - Part of the planned work cannot be completed in accordance with the plan. (medium/medium)	WP6, WP3, WP2, WP1, WP4, WP5	Such a situation is identified early due to regular monitoring. Project management bodies will analyze following possibilities to ensure that planned work can be completed: i) Involve further partners with available resources, ii) Rearrange resources among partners, iii) Ensure involvement of organizations' internal resources, iv) Re-plan work.
4	Project schedule is partly not appropriate (e.g., due to changes) – Common project outcome and impact cannot be achieved as planned. (medium/low)	WP6, WP3, WP2, WP1, WP4, WP5	Project management continuously monitors and measures performed work. Corrective actions (e.g., change of the project plan) will be swiftly performed if necessary. In crucial cases, the coordinator will work on the plan adaptation in close cooperation with EU funding bodies. In extreme cases, like the recent (COVID19) pandemic, the project will continue its operations remotely and if needed extend its duration in order to absorb any problems.
5	Disrupted productivity due to restraining measures. (low/low)	WP6, WP3, WP2, WP1, WP4, WP5	The partners will utilize online services to launch the project and collaborate in undertaking the project tasks.
6	Potential inaccuracies in assumed threat models, undermining the formulation of effective adversarial attacks and hindering the development of the CoEvolution framework. (low/low)	WP3, WP1, WP4	The consortium and the research community have already conducted year-long research on the field, having strong indications that the said threats exist.
7	Integration problems related to the software components of the CoEvolution framework and their deployment. (low/medium)	WP6, WP3	CoEvolution leverages well established technologies allowing natural interoperability of distinct software modules. The proposal leverages successful and consolidated dependability techniques, with tested and validated support provided by the consortium. The proposed ML methods build on ML models with well-established properties supporting risk assessment. Less ambitious techniques will be used to realize intelligent cybersecurity, dependable and privacy- preserving ML functionalities as a proof of concept in case of problems with the proposed methodologies.
8	Delay in delivering the core technological	WP6, WP3, WP4, WP5	Likelihood of the risk is mitigated by the fact that all technical partners bring into the project libraries and components developed as part of previous projects that provide a solid ground

Critical risks & risk management strategy *Grant Preparation (Critical Risks screen) — Enter the info.*			
Risk number	**Description**	**Work Package No(s)**	**Proposed Mitigation Measures**
	components to be integrated in the platform. (medium/low)		upon which to construct the project advanced functionalities. Project impact is mitigated by the fact that the work-plan has a continuous development and integration approach. A usable version (possibly with lesser functionality) will be available at all times. A mockup version of the missing component will be integrated, based on state-of-the-art technology, to allow proof- of-concept testing of the integrated system.
9	Insufficient data for assessing ML components. (low/high)	WP4, WP5	Consortium partners already have preliminary research that suggests there are many data sources that fit the needs of the ML components in CoEvolution. Cooperation among the technical WP ensures that any early issues with data quality and quantity can be fed back to the WP that generate the data. In the unlikely case that not enough suitable data is gathered to fully evaluate the platform, corrective measures will be enacted before the second cycle begins.
10	Difficulty to engage with standardization bodies and WGs for the threat models and protection profiles. (medium/high)	WP2, WP1	Sufficient resources (WP1&2) and partner roles (Innovation Manager and DEM) have been allocated for engaging with the standardization bodies. The project will start very early approaching them and understanding the process and ways to contribute.

PROJECT REVIEWS

Project Reviews *Grant Preparation (Reviews screen) — Enter the info.*			
Review No	**Timing** (month)	**Location**	**Comments**
RV1	18		
RV2	36		